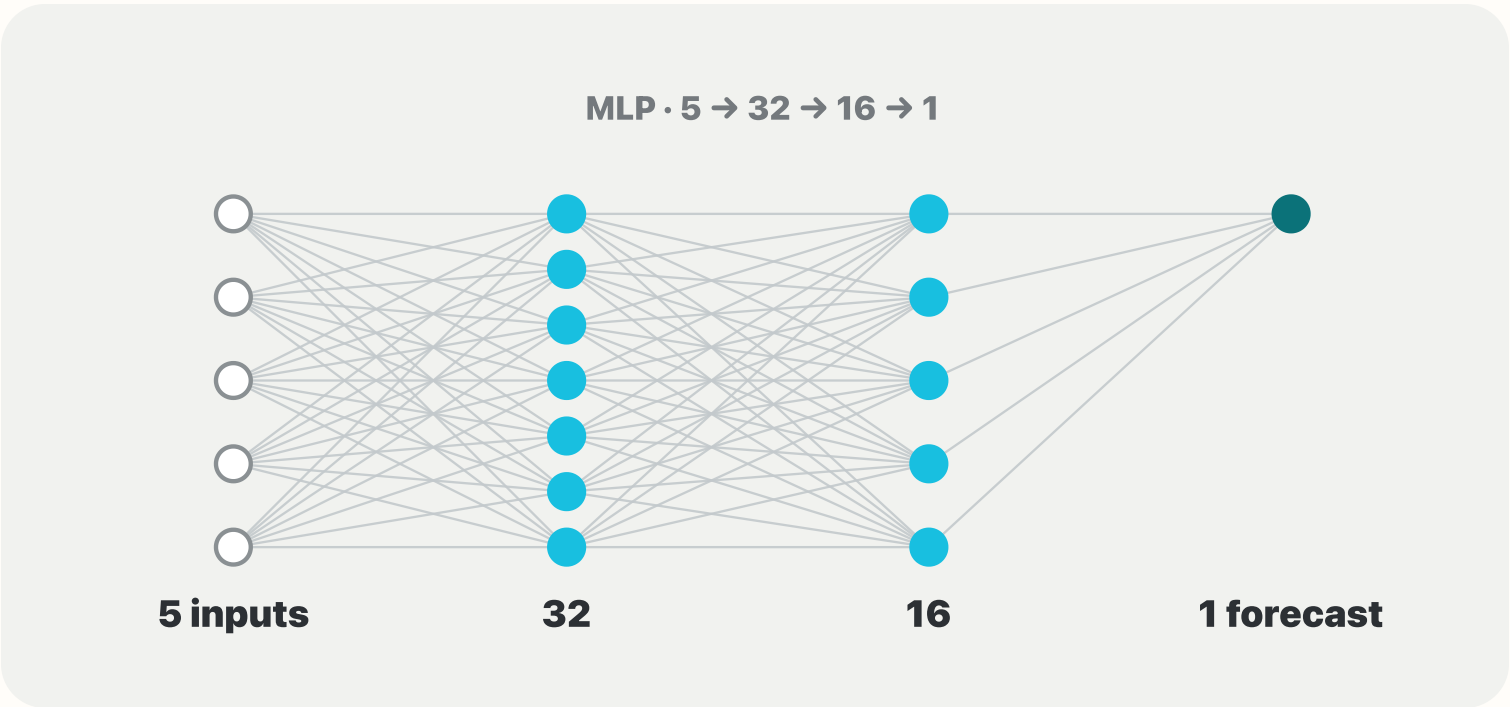


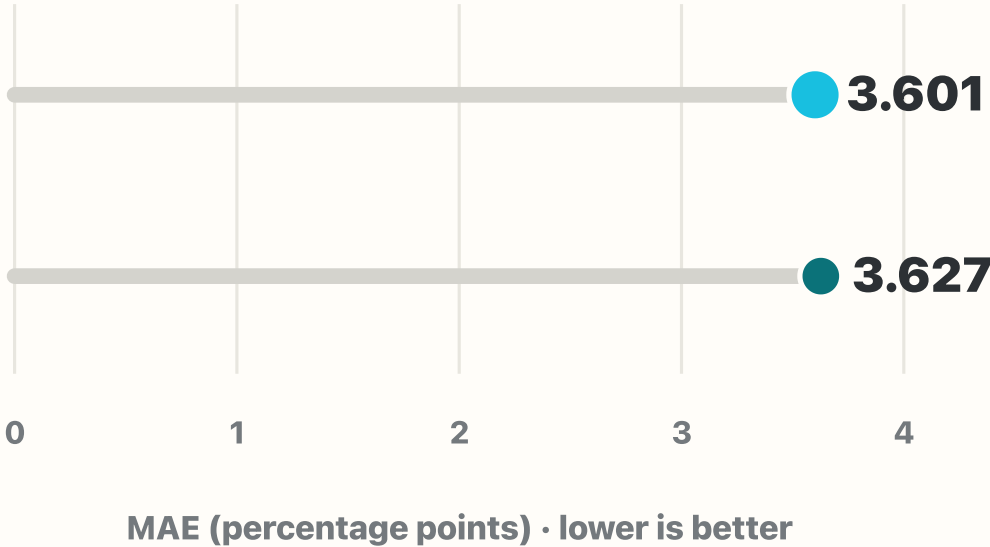
NEURAL NETS LEARN COMPLEX FUNCTIONS. MINE LOST MAE TO A CONSTANT.

TARGET: each bank's percentage change in domestic deposits next quarter.
\$100 million → \$96 million means **−4% actual growth**.
Predict 0%? That is a **4-percentage-point miss**.



0% constant

MLP



THE CONSTANT PREDICTED **0%** GROWTH FOR EVERY BANK.

DEPOSITS CAN LEAVE BEFORE LOANS PAY BACK.

Imagine the internet is down because the AI models broke it 🙄. You walk into your bank and ask for **\$15 in cash**.

YOU: "\$15 CASH, PLEASE." 



CASH IN THE VAULT

\$10



MONEY ALREADY OUT AS LOANS

\$90

\$15 withdrawal. \$10 cash.

**COOL. LET ME JUST TURN PART OF YOUR MORTGAGE INTO
A \$5 BILL.**

SO WHO KNOWS WHAT'S ON THE BALANCE SHEET?



Federal banking regulators collect standardized
Call Reports every quarter.

That gives us bank-by-bank snapshots of cash,
deposits, loans, capital, and more.

CASH

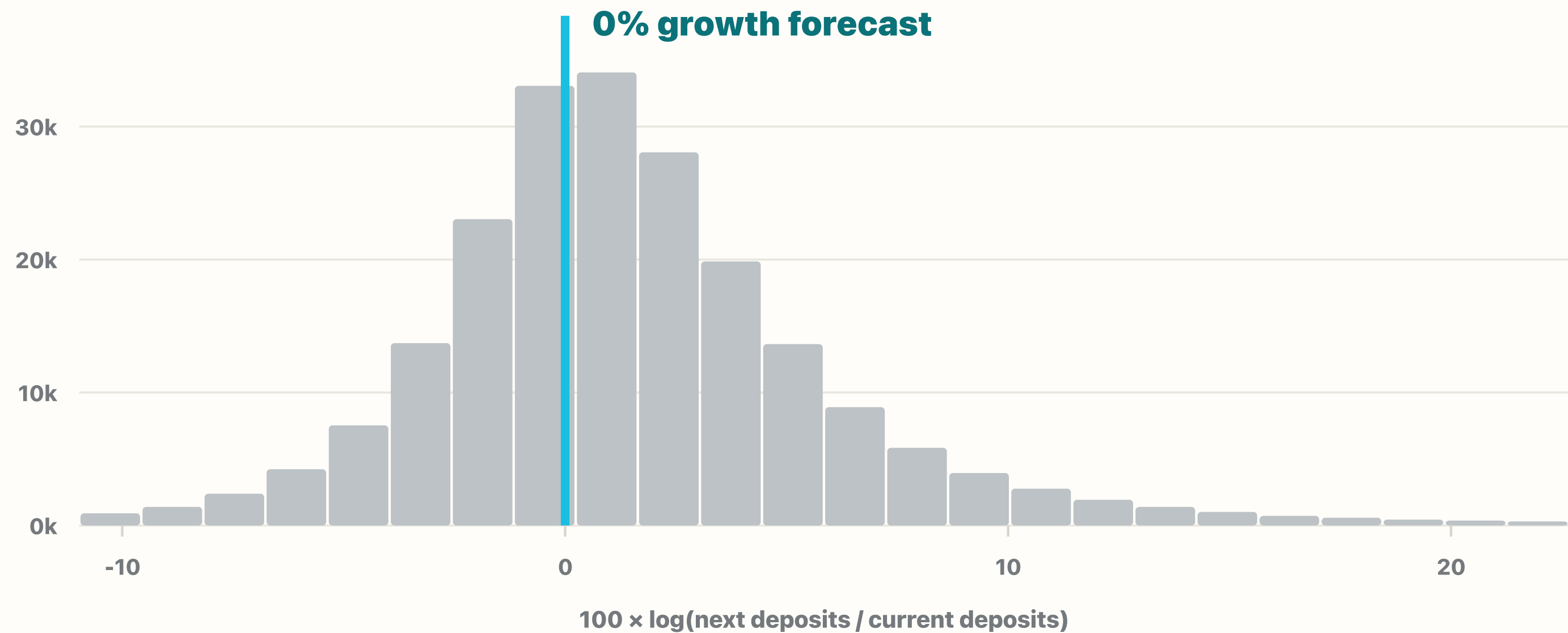
DEPOSITS

LOANS

CAPITAL

THAT'S WHY I CARE ABOUT **WHERE DEPOSITS MOVE NEXT.**

SO WHY WAS 0% GROWTH SO HARD TO BEAT?



A LOT OF THE TRAINING TARGETS LIVE NEAR ZERO.

Central 98% shown · full tails stayed in the experiment · display scale is 100 × log growth

BIG ERRORS GET **SQUARED**. SO THEY CAN PULL THE MODELS AWAY FROM 0%.

A miss 5× as large gets 25× the squared-error penalty.

FIXED BENCHMARK

0% BASELINE

$\hat{y} = 0$

Always predicts 0%.
It never adjusts to reduce a miss.

ILLUSTRATIVE TRAINING ERRORS

2-unit miss $\rightarrow 2^2$

4

VS

10-unit miss $\rightarrow 10^2$

100

BIG MISSES COST MUCH MORE.

LEARNED MODELS

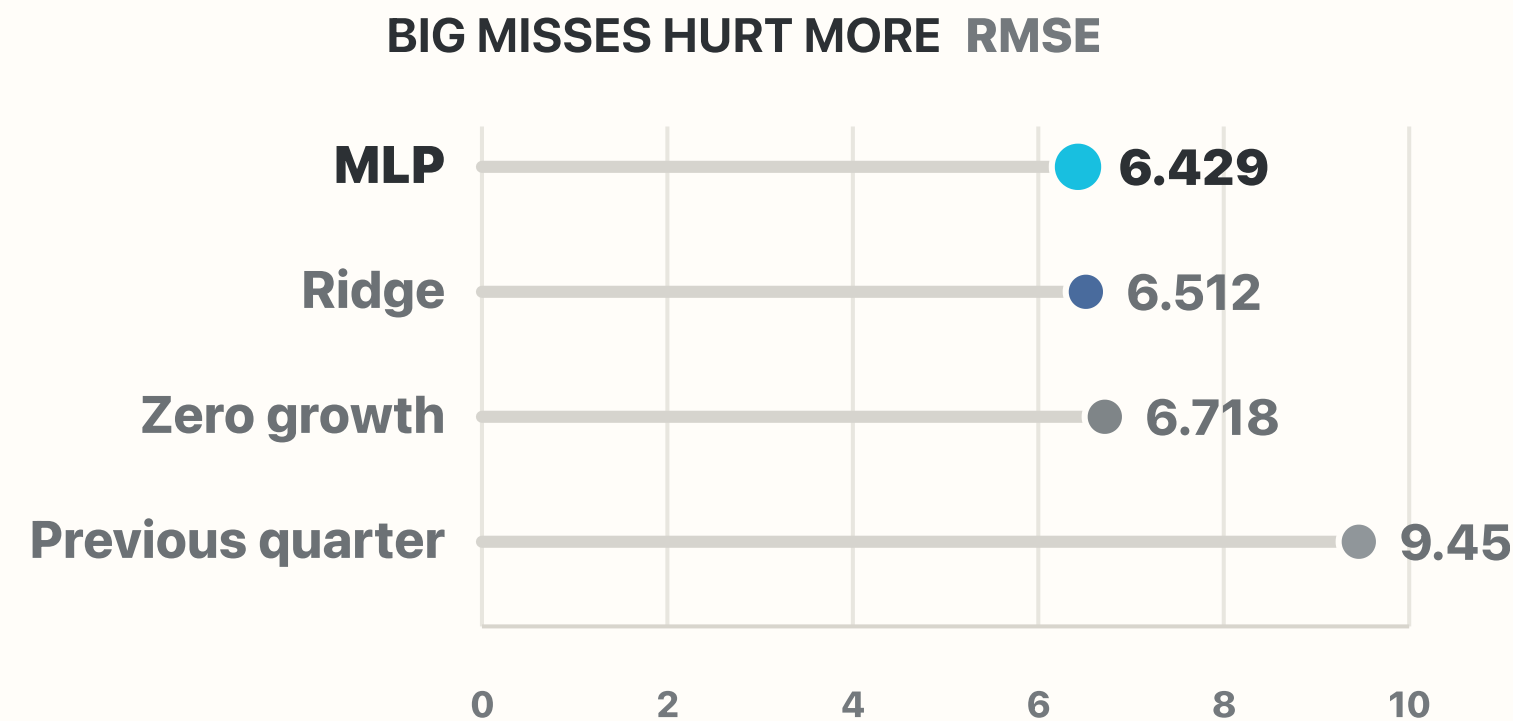
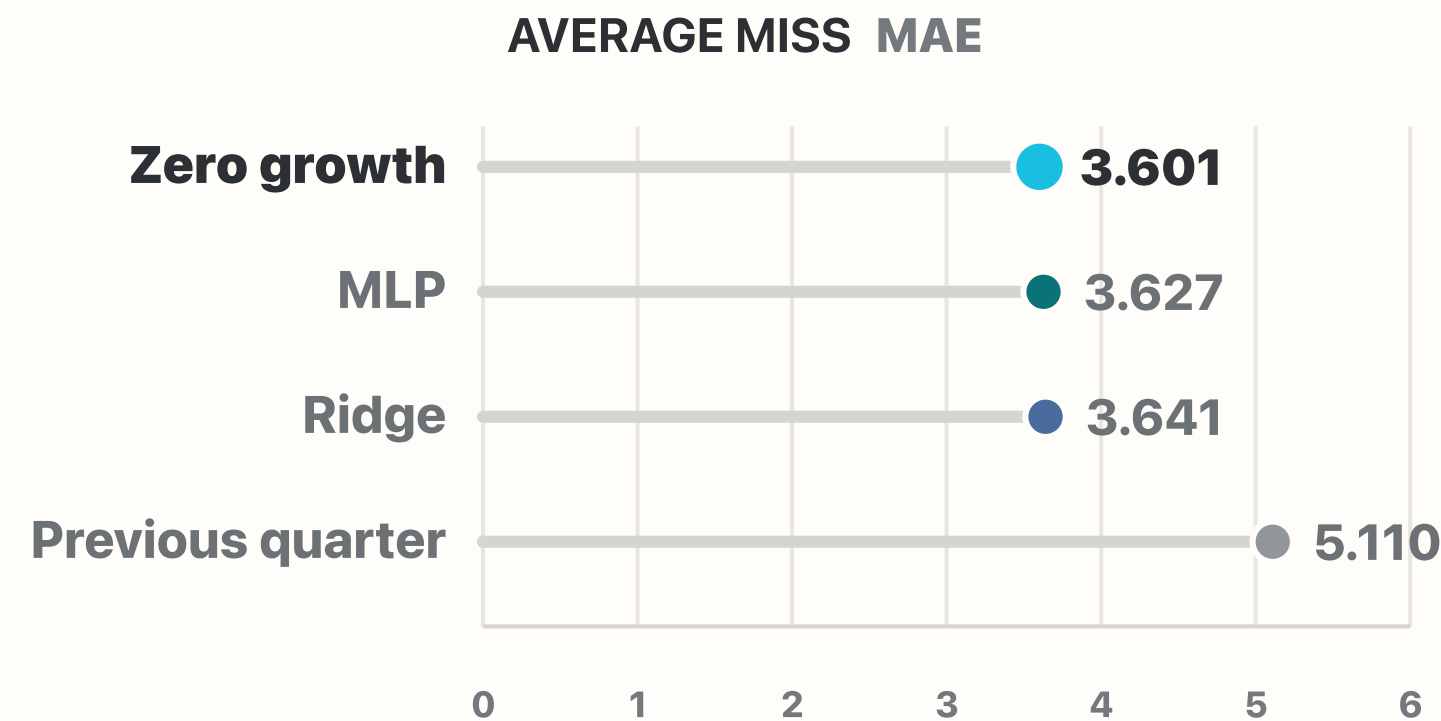
RIDGE + NEURAL NET

Can move away from 0% to reduce those expensive large training misses.

**A LOWER SQUARED ERROR CAN COME WITH
A SLIGHTLY HIGHER AVERAGE ABSOLUTE MISS.**

The example uses log-growth target units. MLP minimizes mean squared error; Ridge adds an L2 penalty. This does not directly optimize the ordinary-growth RMSE shown next.

ZERO WON MAE. THE NEURAL NETWORK WON RMSE.



“WINNER” DEPENDS ON WHAT KIND OF ERROR HURTS.

Same 13,532 evaluation examples · three already-examined 2024 quarters

0% TIES EVERY BANK.

THE LEARNED MODELS TELL ME WHO TO REVIEW FIRST.

Start with the lowest predicted deposit growth. Check the reports and investigate what explains the forecast.

ORDERED REVIEW QUEUE · SIMPLE LINEAR MODEL (RIDGE) · FIRST 10 OF 454

RANK	BANK	PREDICTED GROWTH	0% BASELINE
01	YOUNG AMERICANS BANK	-4.02%	0% · tied
02	PIONEER COMMERCIAL BANK	-3.32%	0% · tied
03	SUMITOMO MITSUI TR BK U SA	-2.83%	0% · tied
04	OLD GLORY BANK	-2.75%	0% · tied
05	TURTLE MOUNTAIN STATE BANK	-2.11%	0% · tied
06	COLUMBUS STATE BANK	-2.05%	0% · tied
07	FDS BANK	-2.03%	0% · tied
08	BOELUS STATE BANK	-1.96%	0% · tied
09	LEMONT NATIONAL BANK	-1.80%	0% · tied
10	CITIZENS NB OF CROSBYTON	-1.75%	0% · tied

HOW MANY OF THE ACTUAL 454
LOWEST-GROWTH BANKS MADE
THE LIST?

Random selection

45.4 expected

Linear model (Ridge)

112 found

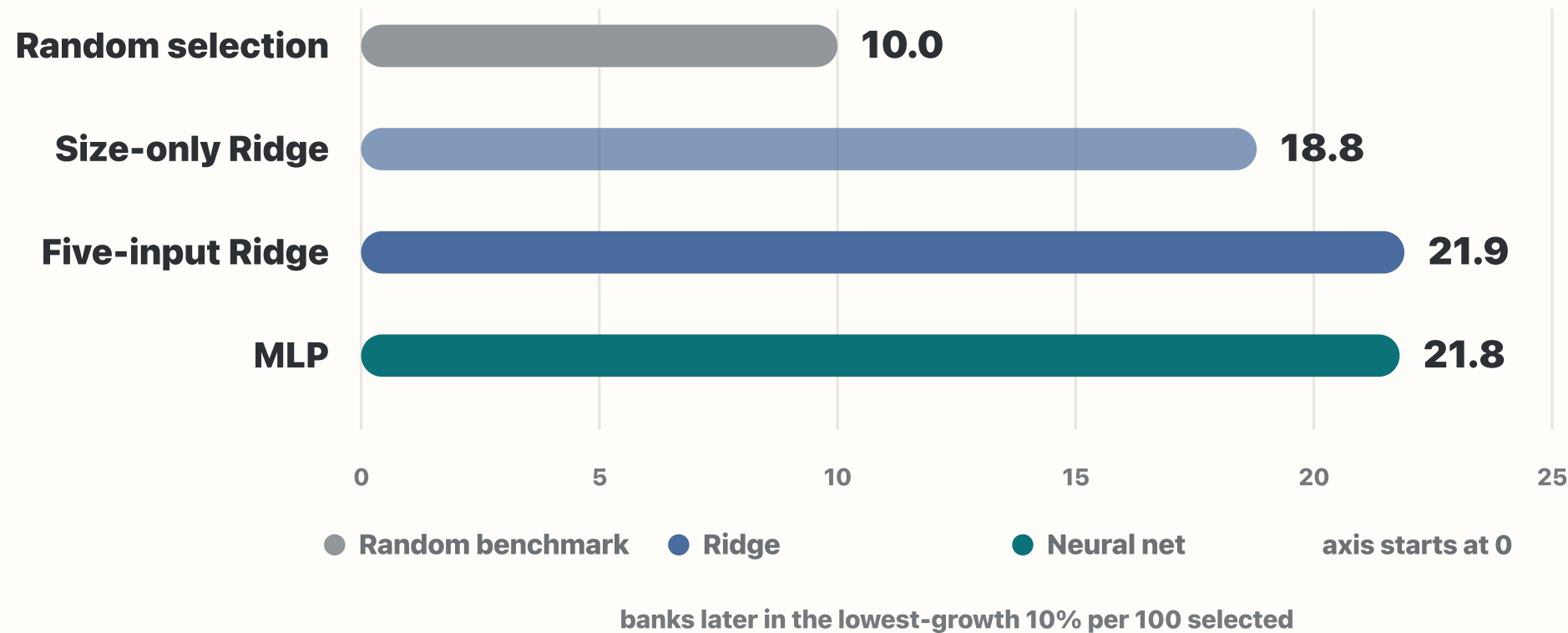
Neural network

109 found

4,536 eligible banks · March → June 2024. “Lowest-growth” = later in that quarter’s bottom 10% of deposit growth.
112 matches across the full linear-model queue; 109 for the neural net. Review priority is not a distress diagnosis.

I THREW OUT 4 INPUTS. KEPT ONLY DEPOSIT SIZE.

How much ranking performance survives with just one clue?
Deposit size = customer deposit dollars. I kept the same linear model and removed the other four inputs.



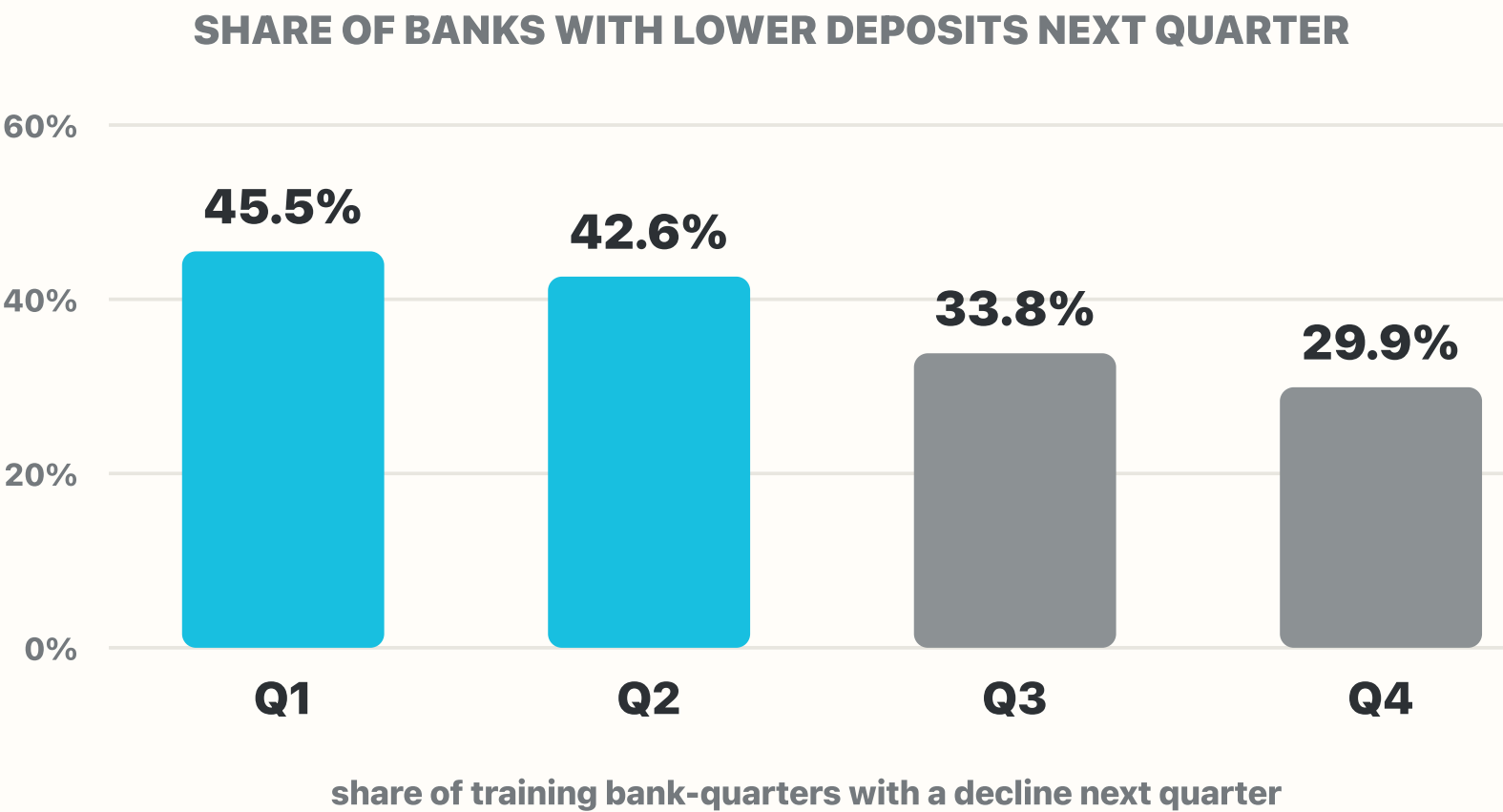
SIZE ALONE: 18.8 FOUND PER 100 REVIEWED.
ALL FIVE INPUTS: 21.9. A GAP OF 3.1.

Equal-quarter averages across three reused 2024 quarters · size-only challenger added after examining 2024 · comparison is not a causal decomposition

DEPOSIT DECLINES WERE MORE COMMON AFTER Q1 THAN Q4.

Q1: 45.5% of training bank-quarters were followed by a deposit decline. **Q4: 29.9%.**

That is how often declines happened, not “banks lost 45.5% of their deposits.”



SO THE QUARTER ITSELF MIGHT BE USEFUL.

Training years only. This is a seasonal pattern in how often deposits declined next quarter.

0% IS STILL THE BENCHMARK TO BEAT. DOES THE CALENDAR RULE EARN ITS PLACE?

June 30 → September 30, 2026 · one forecast per bank · fixed September 22

CALENDAR RULE · Q2 STARTING QUARTER

+0.550%

Same forecast for every bank. If actual growth is 0%, this misses by 0.550 pp.

NO-CHANGE BASELINE

0%

Predict no change. If actual growth is 0%, the miss is zero.

1. WAIT FOR SEPTEMBER REPORTS.

Actual growth =
 $100 \times (\text{September deposits} \div \text{June deposits} - 1)$.

2. MEASURE EVERY MISS.

Absolute miss = $|\text{actual growth} - \text{forecast}|$.
Example: +1.00% actual → 0.450 pp calendar miss; 1.000 pp zero miss.

3. COMPARE THE AVERAGES.

Lower MAE wins this quarter.
If the calendar rule loses, report the loss. Inspect misses before changing the rule.

**IF DEPOSITS DON'T CHANGE, ZERO IS EXACT.
THE CALENDAR RULE HAS TO EARN THE IMPROVEMENT.**

As of Sep 22, 2026: June data are listed in FDIC BankFind; the Sep 30 quarter has not ended. Saved rule = Q2 training median (+0.550399842%); developed after seeing 2024. Score matched banks with valid positive June and September domestic deposits; disclose missing reports and merger cases. Example is illustrative. One quarter is not lasting proof.